MIL-S-52699B(ME)
30 August 1976
SUPERSEDING
MIL-S-52699A(ME)
31 October 1972

MILITARY SPECIFICATION

SNOW-REMOVAL UNIT, ROTARY; TRUCK-MOUNTED, DIESEL-ENGINE-DRIVEN 1500 TONS-PER-HOUR CAPACITY

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 36,000-pound gross vehicle weight rating (GVWR) diesel-engine-driven carrier equipped with a diesel-engine-driven rotary snowplow head capable of removing and casting snow at the rate of 1,500 tons per hour.

1.2 Classification. The snow-removal units shall be of the following classes, as specified:

Class A - Nonwinterized - operational to minus 25°F.
Class B - Winterized - operational to minus 50°F.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 3825

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SPECIFICATIONS

FEDERAL

O-A-548	- Antifreeze/Coolant, Engine: Ethylene Glycol, Inhibited, Concentrated.
UU-T-81	- Tags, Shipping and Stock.
VV-F-800	- Fuel Oil, Diesel.
ZZ-T-381	- Tires, Pneumatic, Vehicular (Highway).
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-1283	- Can, Gasoline, Military, 5-Gallon.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-L-6085	- Lubricating Oil: Aircraft Instrument, Low Volatility.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-L-10324	- Lubricating Oil, Gear, Sub-Zero.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-A-11755	- Antifreeze, Arctic-Type.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.

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STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS51118	- Pintle Assembly, Towing: 40,000 Pound Capacity, Manual Release.

FEDERAL

FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment.
FED. STD. No. 595	- Colors.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

TECHNICAL INFORMATION PACKAGE

(Application for copies should be addressed to USA Mobility Equipment Research and Development Command, ATTN: DRXFB-H, Fort Belvoir, VA 22060.)

AMERICAN WELDING SOCIETY, INC.

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

DEPARTMENT OF TRANSPORTATION (DOT)

49 CFR 567 and 568 - Certification.
49 CFR 325, 390, and 393 - DOT Federal Motor Carrier Safety Regulations (FMCSR).
49 CFR 571 - DOT Federal Motor Vehicle Safety Standards (FMVSS).

ENVIRONMENTAL PROTECTION AGENCY

EPA, 45 CFR, Part 1201 - Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC.

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

3. REQUIREMENTS

3.1 Description. The snow-removal unit (hereinafter called "snowplow") shall include but not be limited to: (1) A cab-forward, four-

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wheel-drive, front-wheel-steer carrier with the diesel carrier engine mounted at the rear or under the cab; (2) a diesel engine for the rotary plow mounted behind the cab; and (3) a hydraulically controlled rotary snowplow-head mounted on the front of the vehicle. Each snowplow shall be furnished as a complete self-contained unit with all parts and accessories necessary for operation. The weight of the snowplow including carrier, rotary plow, tools and equipment, fuels and fluids, and an allowance of 350 pounds for operating personnel shall not exceed 36,000 pounds.

3.2 First article (preproduction model). The contractor shall furnish a snowplow for examination and testing within the time frame specified (see 6.2), to prove prior to starting production, that his production methods will produce snowplows that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4, and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests.

3.3 Material. Material shall be as specified herein. Materials not specified shall be those used for the same purpose on a snowplow of comparable size regularly sold on the commercial market and shall be subject to all provisions of this specification. Components, particularly of the electrical, fuel, and exhaust systems, and brake air lines shall resist harmful effects of dust, water, rain, and corrosion. To the maximum practicable extent, materials and treatments shall be utilized to provide resistance to corrosion.

3.3.1 Threaded parts. Threaded parts shall be in accordance with Handbook H28.

3.4 Environmental conditions. The snowplow shall be capable of starting and shall meet the operational chacteristics specified herein under any of the following conditions or combination of conditions.

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3.4.1 Applicable to Class A.

(a) Temperature conditions. The carrier shall operate in any ambient temperature from plus 115°F to minus 25°F, and the rotary unit shall be capable of satisfactory operation in any ambient temperature from plus 50°F to minus 25°F. The complete unit shall not be damaged by storage in any ambient temperature from plus 155°F for 4 hours per day to minus 65°F for periods of 72 hours. Certification of compliance with storage requirements will be acceptable.

(b) Elevation conditions. The carrier and rotary unit shall operate at any elevation from sea level (Bar. 29.92 in, Hg.) to 8,000 feet.

3.4.2 Applicable to Class B.

(a) Temperature conditions. The carrier and rotary unit shall operate as specified in 3.4.1(a), except that the starting and operating temperature shall be minus 50°F accompanied by a 5 mph frontal wind. Satisfactory operation after starting shall not be dependent on the application of heat from the winterization system to the engines or to the power trains.

(b) Elevation conditions. The carrier and rotary unit elevation requirements shall be as specified in 3.4.1(b).

3.5 Safety and human factors.

3.5.1 Safety. Rotating parts (except the snow feed and discharge mechanism), reciprocating parts, electrical components, and parts subject to high operational temperatures that are so located as to become a hazard to the safety of operating or maintenance personnel shall be insulated, enclosed, or guarded to the extent necessary to eliminate the hazard.

3.5.1.1 Walking surfaces, steps, and handholds. Platforms, catwalks, steps, and other walking or standing surfaces of the snowplow shall be of open metal grating with nonskid surface. Catwalks, ladders, steps, and handholds shall be provided on the sides of the snowplow cab and superstructure so that access thereto may be unhampered and nonhazardous. All walking surfaces, steps, and handholds shall be in accordance with SAE J185.

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3.5.1.2 Safe operating conditions. The controls of the snowplow shall provide safe operating conditions for protection of operating personnel, of the snowplow, and of materials and equipment close to the path of the snowplow during plowing, loading, spot casting, forward and reverse travel, and any other function controlled by the operator. All hand-operated controls, except steering, which may require attention during operation or travel shall be located for operation by one hand or the other, but not by both, in order to provide uninterrupted control of the steering by one hand.

3.5.1.3 Neutral start, carrier engine. An attachment or control mechanism shall be provided within, or connected to, the shift linkage to prevent engagement of the carrier power train while the engine is being started.

3.5.1.4 Neutral start, plow engine. An attachment or control mechanism shall be provided within, or connected to, the power train shift linkage or clutch to prevent rotation of the plow while the plow engine is being started.

3.5.1.5 Reflectors. Reflectors conforming to SAE J594 shall be furnished and mounted on the carrier in accordance with DOT Motor Carrier Safety Regulations.

3.5.1.6 Air horn. An air horn assembly shall be provided. The air horn shall be positioned and located so that it will not become inoperative due to blowing snow.

3.5.2 Human factors. The characteristics of the snowplow shall provide for operation by personnel ranging from the small man (5th percentile) clothed, through the large man (95th percentile), arctic clothed in accordance with SAE J833 and SAE J925.

3.5.2.1 Sound level. The sound level of the snowplow shall not exceed 88 dB(A) when measured in accordance with SAE J88. With the cab windows, doors, and vents in the fully closed position the sound level of the snowplow shall not exceed 90 dB(A) when measured in accordance with SAE J919. If the sound level of the snowplow exceeds 85 dB(A) when measured in accordance with SAE J919, a warning plate or decal containing the following shall be permanently affixed to the snowplow in a conspicuous, protected location in the operator's compartment:

"WARNING, EAR PROTECTION REQUIRED WHEN OPERATING THIS SNOWPLOW."

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3.6 Ease of maintenance. The snowplow shall be constructed in such a manner that major assemblies are accessible for repair or replacement with a minimum removal or disturbance of adjacent parts or components and by use of general-purpose tools. Operating parts and accessories shall be accessible for regular maintenance operations. Drain outlets shall be located for easy access. Provisions shall be made to permit the use of receptacles for collecting the drainage from the radiator, crankcase, and other components. Covers or plates which must be removed for component adjustments shall be equipped with quick-disconnect fastenings. Ease of maintenance provisions shall incorporate, to the maximum practicable extent, features providing ample clearances for maintenance and servicing operations at low temperatures by personnel wearing heavy gloves and hampered by bulky clothing and foot gear. Intricate locking devices and controls with threaded fastenings which can be over-torqued by operators lacking feel through thick gloves or numbness shall be avoided. Dimensions of hand access openings shall be in accordance with SAE J925.

3.7 Performance.

3.7.1 Production. The snowplow shall cast snow weighing between 20 and 40 pounds per cubic foot at a rate of not less than 1,500 tons per hour, Under no-wind condition, the snow shall be cast a distance of at least 125 feet to the right or left of the snowplow while a strip at least 8-1/2 feet wide is cleared in snow from 1 to 4 feet deep with the snowplow operating at a continuous forward speed and removing snow at the rate of 1,500 tons per hour. The casting distance is defined as the distance from the centerline of the carrier chassis to the center of mass of the discharged snow pattern.

3.7.2 Plow head operation. The discharge chutes shall be adjustable for operation through a horizontal arc of not less than 200 degrees (100 degrees to the right and 100 degrees to the left) and shall spotcast the snow throughout the range of 14 to 60 feet from the center of the chute. The plow cutting mechanism shall give equally satisfactory performance when clearing snow from 1-1/2 inches to 4 feet deep leaving not more than 1/2 inch of residual snow. Except for shoe or caster moldboard adjustment, any adjustments necessary to accomplish this performance shall be controllable from inside the cab during operation. Provisions shall be made for the installation of cutters (see 3.10.8) to equip the snowplow for 8-foot-deep snow removal.

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3.8 Dimensions. The snowplow shall comply with the following dimensional requirements:

(a)	Overall height excluding plow head, lights, and exhaust stacks.	- 132 inches maximum
(b)	Overall width without plow head and side mirrors.	- 101 inches maximum
(c)	Wheelbase.	- 160 inches maximum
(d)	Vehicle clearance circle without plow head (SAE J695).	- 75 feet maximum
(e)	Ground clearance.	- 12 inches minimum
(f)	Pintle-to-ground height.	- 30 inches minimum 36 inches maximum

3.9 Carrier. The carrier shall conform to all applicable safety requirements of DOT Federal Motor Vehicle Safety Standards (FMVSS) and Federal Motor Carrier Safety Regulations (FMCSR). The carrier shall have a gross vehicle weight rating (GVWR) of not less than 36,000 pounds in accordance with DOT certification.

3.9.1 Diesel engine. For the purpose of selecting the engines, the snowplow shall be classified as a Class II end item in accordance with MIL-STD-1410. The engine shall be diesel. The engine, including all systems components and accessories, shall conform to MIL-STD-1410, Class II, except as otherwise specified herein. The engine shall be capable of operating on fuels and lubricating oils as specified in MIL-STD-1410. The engine shall comply with the Environmental Protection Agency Regulations governing control of air pollution for new motor vehicles and new motor vehicle engines, in effect on the date of manufacture. The engine shall meet all operational requirements and conditions specified herein without exceeding any of the operating limits specified in Table I of MIL-STD-1410.

3.9.2 Cranking motor. The engine shall be furnished with a heavy-duty 12-volt cranking motor. The cranking motor shall be furnished by or approved by the engine manufacturer.

3.9.3 Battery charging system. The engine shall be furnished with a 12-volt negative ground battery charging system. The battery charging system shall be of the waterproof brushless alternator type of not less than 90 ampere capacity. The battery charging system shall be furnished by or approved by the engine manufacturer.

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3.9.4 Low temperature starting system. A low temperature starting and priming system in accordance with MIL-STD-1410 shall be furnished and shall be electrically (solenoid) operated. The priming system switch shall be located on the cab instrument panel and shall be clearly and permanently marked as to its function.

3.9.5 Engine speed control system. The engine speed control system shall use a foot treadle and shall be located in a position so that it may be operated by the carrier operator's right foot. A hand throttle is not required for the carrier engine.

3.9.6 Air induction system. The engine shall be equipped with a heavy-duty air cleaner and a restriction indicator in accordance with MIL-STD-1410. The restriction indicator shall be mounted on the cab instrument panel and shall be clearly and permanently marked as to its function.

3.9.7 Exhaust system. An exhaust system complete with exhaust pipe, muffler, and tailpipe shall be provided. When the engine is rear-mounted, the end of the tailpipe shall be in a horizontal position above the engine housing aimed toward the rear with the tip cut to restrict entrance of rain and snow. With a cab-over-engine configuration, a vertical-stack exhaust with raincap shall be furnished. The horizontal section of the pipe shall slope slightly downward. All components, except the exhaust manifold shall be fabricated of aluminized or stainless steel.

3.9.8 Radiator winterfront. Thermostatically controlled radiator shutters shall be supplied. Shutters shall fail-safe in the open position. Shutters and linkages shall be corrosion-resistant.

3.9.9 Radiator. The radiator shall be matched to the engine and drive-train heat rejection load.

3.9.10 Carrier drive train. The drive train shall transmit the maximum available torque and horsepower, and shall resist the shocks imposed by all operations specified herein without damage or permanent deformation. Oil temperatures in drive train components shall not exceed 250°F during any operation specified herein.

3.9.10.1 Torque converter. A torque converter shall be furnished and shall provide a torque multiplication ratio at stall speed within the range of 2.2 : 1 to 3.3 : 1. Heat-exchanging components shall maintain the required operating temperature throughout the ambient temperature

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range specified in 3.4 at all vehicle loads and speeds when adequate gear ratios are selected which permit the converter to operate between 70 percent and 100 percent efficiency. When the converter is subjected to 30 seconds of wide-open-throttle stalling, the oil temperature shall not exceed 250°F from an initial temperature of 130°F and at an ambient temperature of not less than 30°F.

3.9.10.2 Main and auxiliary transmissions. The carrier shall be equipped with a full-torque powershift constant mesh main transmission, and an auxiliary transmission, if required. The main transmission shall be capable of being shifted while the maximum torque of the engine multiplied by the torque converter is being delivered and shall include not less than five forward ratios or four foward ratios combined with at least a two speed auxiliary transmission and not less than one reverse ratio. To facilitate rocking the vehicle back and forth the time for full engagement shifting between forward and reverse shall not exceed 2 seconds. The main transmission shall be equipped with a downshift inhibitor system that prevents driver shift control action from overspeeding or damaging the engine, transmission, or drive train. The transmission shift pattern shall prevent the driver from inadvertently shifting into reverse. The controls shall permit rapid intentional shifting from forward to reverse and from reverse to forward. Heat-exchanging components shall maintain the required lubricant temperature range throughout the ambient temperature range specified in 3.4 at all vehicle loads and speeds when adequate gear ratios are selected which permit the converter to operate between 70 percent and 100 percent efficiency. Shift lever positions for the auxiliary transmission shall be clearly marked. When the auxiliary transmission cannot be shifted with the vehicle in motion, a warning plate shall be installed near the auxiliary transmission control lever visible to the carrier operator. The vehicle shall be towable with the engine inoperative without damaging the transmission(s), and without disassembly other than drive shaft universal joints.

3.9.10.3 Final drive. The drive train shall furnish power to all four wheels so that slippage of one or more wheels will not decrease transmission of power to the wheel or wheels having traction.

3.9.11 Chassis. The carrier chassis shall be furnished with four single wheels, four-wheel drive, amd front-wheel steering. A rear bumper of steel channel shall be furnished, and shall have a length of not less than three-fourths the overall width of the carrier.

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3.9.12 Snowplow speed and drawbar pull. The fully equipped snowplow shall maintain the forward and reverse speeds and the drawbar pull listed in Table I. This performance shall be obtained without exceeding a wheel slippage (travel efficiency) of 15 percent.

Table I. Vehicle Speed and Drawbar Pull

Direction	Speed (mph)	Minimum drawbar pull (lb)
Forward	1	50 percent of GVW
Forward	6	21 percent of GVW
Forward	10	13 percent of GVW
Forward	15	7 percent of GVW
Forward	20	4.5 percent of gvw
Forward	35	2.1 percent of GVW (calculated)
Reverse	8	No requirement

3.9.13 Pintle assembly. A medium-duty, nonrotary pintle assembly conforming to MS51118 shall be centrally located on or beneath the rear bumper. The pintle mounting shall be suitably reinforced to sustain the capacity of the pintle assembly.

3.9.14 Weight distribution. The weight of the complete snowplow shall be distributed to provide not less than a 9,000-pound load on the front wheels with the plow head resting on the ground and not greater than a 20,000-pound load with the plow head raised in the travel position.

3.9.15 Tires. Tire size shall be not less than 14.00 x 20, 18-ply rating. Tires shall have a block or diamond tread pattern and shall conform to ZZ-T-381. All tires shall be of the same size, type, ply rating, and tread pattern. Ozone-resistant additives shall be used in the tires. Clearance shall be provided for the mounting of tire chains on all wheels in accordance with SAE J683. Four tire chains shall be furnished and shall be stored in the toolbox of the snowplow. The tires shall conform to the recommendations of the Tire and Rim Association Yearbook in regard to loading, tire pressure, and speed. The contractor shall furnish a certification from the tire manufacturer that the tires furnished conform to the recommendations of the Tire and Rim Association Yearbook in regard to loading, tire pressure, and speed and are suitable for the application.

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3.9.16 Steering. The carrier shall be equipped with front wheel steering controlled by the carrier operator seated on the right side of the cab. The operator's effort required on the rim of the steering wheel, from lock to lock position, shall not exceed 18 pounds under static conditions or 5 pounds under vehicle movement when operated on dry concrete. Emergency steering control shall be provided such that the snowplow may safely be maneuvered with a dead engine when tested as specified in 4.5.2.8.

3.9.17 Brake system. The carrier shall be equipped with a weatherproof, straight airbrake system that shall comply with the requirements of the California Vehicle Code and the DOT FMVSS. All brake-system lines and components shall be located, installed, and protected so that damage from ground obstructions and debris is effectively prevented and that maximum efficiency and ease of maintenance is obtained.

3.9.17.1 Compressor. The compressor shall be water-cooled, engine-driven, and engine-oil-lubricated with a rating of not less than 12 cfm. The compressor shall have an automatic pressure control with a factory setpressure range of 95 to 120 psi. The compressor shall draw filtered air from the engine air cleaner manifold through a check valve.

3.9.17.2 Brake components. Components shall comply with 3.3, modified when required to include the following:

(a) Alcohol evaporator.
(b) Wet and dry reservoirs each with safety valves and drain cocks.
(c) Front axle limiting and quick-release valve.
(d) Emergency stopping system.
(e) Parking brake system. With the snowplow on a level concrete surface, the parking brake shall hold the vehicle when a force equal to 50 percent of the weight of the snowplow is applied to either end of the snowplow. The parking brake shall be applied and released by a hand or a foot control located in the operator's compartment. The parking brake control shall require not more than a 90-pound push or pull to meet the above requirement.
(f) High-capacity, foot-operated service brake application valve.
(g) Stoplight switch.
(h) Anti-lock system in accordance with DOT FMVSS.

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3.9.18 Electrical system. The snowplow shall be equipped with a 12-volt, waterproof, vehicular electrical system for travel operations, for floodlighting the work area at night, for slave starting the engines, and for charging the batteries and starting the engine. Except as specified herein, the electrical system shall be in accordance with SAE Handbook, Electrical Equipment and Lighting Section, and the DOT requirements. All lights and reflectors shall conform to the SAE Handbook. All lights and reflectors shall be located in accordance with the DOT requirements with the exceptions and additions specified in 3.9.18.1.

3.9.18.1 Electrical components. Components shall comply with 3.3 modified when required to include the following:

(a) Headlights, one on each side of cab roof.
(b) Front marker lights.
(c) Two sealed-beam spotlights, 50-watts each, one on each side of the cab near the top, with direction remote-controlled from the seats.
(d) Horizontally rotating red warning light, worm-gear driven, not less than 35,000 candlepower, mounted on cab or plow engine housing.
(e) Turn signal system, including a hazard warning signal flasher.
(f) Two rear-aimed floodlights for illumination while backing and turned on by shifting transmission into a reverse gear.
(g) Batteries, MS35000, size 6TN.
(h) Cab dome light.

3.9.18.2 Battery box. A battery box shall be furnished. Interior size for a set of batteries shall provide a clearance in excess of 1 inch between any part of the cover and battery terminal lugs, posts, or other battery components. The interior size shall provide a minimum clearance of 1 inch around the perimeter of each set of batteries. The box shall be constructed of steel, except the battery box cover may be constructed of a lighter material, and shall be treated with an acid-resisting coating on the interior. Each pair of batteries shall be held in place by a retainer with flat washers and wingnuts. The box shall be provided with drain holes.

3.9.18.3 Wiring. The wiring shall be fabricated and installed as recommended in SAE J555, except as specified herein. Where two or more

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wires follow the same routing, the wires shall be formed into harnesses. Battery cables shall satisfy the SAE starter voltage drop for heavy-duty diesel use and current requirements, shall be as short as possible, and shall not be spliced. All wires shall be number or letter coded near each connection. Friction retainment connectors shall be lubricated prior to assembly and each connector assembly shall be secure. Removal of major assemblies such as engine, cab, and instrument panel shall be facilitated by disconnecting threaded plugs and receptacles for all circuits in each assembly. Terminal-type connectors may be used on the floodlight system.

3.9.18.4 Slave receptacle. A slave receptacle shall be installed near the batteries, and connected to permit engine slave starting and battery charging. A plug to fit the receptacle, shall be furnished, suitable for attaching to a slave cable.

3.9.19 Walkways and steps. The walkways shall provide access on both sides of the carrier from the rear bumper to the cab doors, and shall be provided with outside handrails.

3.9.20 Cab. The cab shall be fabricated of sheet steel. The cab shall conform to and include the following:

(a) Fully enclosed type, overall width not less than 72 inches, not including handles or rear-view mirrors.
(b) Right-hand drive (carrier controls on the right-hand side).
(c) All glass shall be shatterproof and tinted and shall conform to SAE J673.
(d) Flat, two-pane with vertical center divider, windshield designed with a reverse rake (angled between 5 to 25 degrees forward from the bottom measured from the vertical plane). Windshield glass shall be semitempered, laminated plate glass.
(e) Two hinged side doors not less than 30 inches wide by 50 inches high. Both doors shall be equipped with heavy-duty, hold-open, friction catches.
(f) Two or three rear windows.
(g) Windows in the side doors shall be actuated by a lever-operated control or handcrank and shall be readily opened and closed by personnel wearing heavy gloves.
(h) Two cushioned sun visors equipped with suitable locks or mountings to prevent position changes due to vibration.

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(i) Two cushioned, integral seat-and-back framed seats with padded arm rests both with foam-rubber or foam polyurethane padding. The driver's seat shall be adjustable up and down (3-inch minimum travel), and fore and aft (4-inch minimum travel). Seats shall be covered with abrasion-resistant coated material weighing not less than 32 ounces per square yard. Each seat shall be provided with seat belts conforming to SAE J4 mounted in accordance with SAE J787.

(j) Cab heater, defroster and insulation as specified in 3.9.21 and 3.9.22.

(k) Two windshield wipers equipped with shaft extensions or other means permitting manual operation from within the cab.

(l) Two hinged and screened cowl vents in the front cab sheet metal directing ventilation to the floor boards.

(m) Right and left rectangular, outside, rearview mirrors. Each mirror shall have an area of not less than 90 square inches and shall be solidly supported by not less than two adjustable supporting arms. Both mirrors shall be easily folded against the cab or doors without the use of tools. Both mirrors shall be provided with electric heating to remove snow and ice. The heating circuit control shall be mounted on the instrument panel and shall include a green indicator light.

(n) Handgrips and steps of ample size for ise of personnel wearing heavy gloves and arctic thermal boots mounted on the cab exterior.

(o) Interior paneling and insulation.

3.9.20.1 Instrument panel. The panel shall contain two groupings of instruments, indicators, and controls; one grouping for each engine. The instrument grouping for the plow engine shall be located to the left of the steering column, near the centerline of the cab and visible from the plow operator's seat. The instrument grouping for the carrier engine shall be visible from the driver's seat. A toggle switch shall be provided beneath the fuel indicator to provide a choice of fuel tank readings. Fasteners shall permit removal of instruments, clusters, or subpanels from the face of the panel assembly to allow removal or repair of any gage, indicator, or control and to allow access to the wiring. All gages, indicators, and controls shall be identified. The instruments shall be of such quality and so mounted as to function under all conditions specified herein. All instruments and controls shall be provided with indirect nonglaring illumination.

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3.9.20.1.1 Carrier instruments. The carrier instrumentation shall be in accordance with 3.3, modified when required to include the following:

(a) Speedometer.
(b) Odometer.
(c) Engine oil pressure gage.
(d) Engine coolant temperature gage.
(e) Torque converter oil temperature gage.
(f) Fuel gage.
(g) Air pressure:

(1) Gage.
(2) Low pressure warning light (red).
(3) Low pressure buzzer.

(h) Voltmeter or ammeter.

3.9.20.1.2 Plow instruments. The plow engine instrumentation shall be in accordance with 3.3, modified when required to include the following:

(a) Tachometer.
(b) Hourmeter.
(c) Engine oil pressure gage.
(d) Engine coolant temperature gage.

3.9.21 Heater and defroster. An engine coolant, fresh-air-type heater(s) with defroster shall be furnished, with a rating of not less than 30,000 Btu/hr. Windshield defroster outlets shall be of the louver type extended not less than 60 percent of space width to be heated. Controls shall be provided for variable selection of fresh air volume, fresh or recirculated air, floor or defroster outlet air, blower speed, and heater temperature (coolant volume). The controls shall be grouped in the same general location, primarily accessible to the driver. The heater installation shall be supplied with drain valves and, if necessary, air lock bleed valves. All exposed heater components and piping shall be protected against damage. The fresh-air intake shall be located so as to restrict the entry of snow. Window defrosters and fans shall maintain clear vision through 75 percent of windshield and side window areas.

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A ducting system shall be provided for the defrosters and to direct and distribute heat to the operators. The heater shall maintain a cab temperature of not less than 35°F with an ambient temperature of minus 25°F for Class A snowplows and minus 50°F for Class B snowplows. Separate rubber-blade fans shall be installed to direct streams of air over the rear windows.

3.9.22 Cab insulation. The floor of the cab shall be insulated with 1/4-inch thick felt, neoprene-coated, floor mat. The floor mat shall be sectionalized to facilitate removal and installation. Sections shall be framed on all edges with fastened, metal threshold strips extending at least 2 inches over the edges of the floor mat pieces to prevent curling or tearing of mats. The top and sides of the cab shall be insulated with mastic-type insulation or polyvinyl chloride sheet insulation not less than 1/4-inch thick. All insulation shall be fire retardant. Doors and other openings shall have a weather-stripping seal. Weather stripping for sealing the bottom of the door areas shall be installed along the bottom edges of the doors. Levers and controls shall be equipped with boots, or an equivalent seal, where they pass through the floor or wall.

3.9.23 Controls. Operating controls including those for the chutes, shall be located in the carrier cab and shall be suitable for operation by personnel wearing heavy gloves and arctic clothing. The plow controls shall be located in the cab so that they may be operated by either the carrier driver or the rotary plow operator.

3.10 Rotary plow. The rotary plow components shall include a diesel engine, power train, rotary plow head, and hydraulic control system.

3.10.1 Plow engine. The engine, including all systems, components and accessories shall conform to MIL-STD-1410 and as specified in 3.11.1. Engine speed shall be controlled by a hand throttle. The throttle shall be a precise-control, rotating-knob type, for adjusting and holding (within 150 rpm) the setting at any engine speed from idle to governed speed. A quick-release mechanism shall return the hand throttle control to idle position. The engine shall be turbocharged.

3.10.2 Safety devices. The engine shall be provided with safety devices in accordance with MIL-STD-1410 that will automatically shut off the engine under conditions of high coolant temperature, low lubricating oil pressure, or overspeed, in accordance with the recommendations of the engine manufacturer. An operator override shall be furnished after shut-off.

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3.10.3 Power rating. The engine, as installed, shall develop not less than 300 brake horsepower at the SAE J816 standard ambient conditions and shall be certified and warranted by the engine manufacturer as being suitable for the intended application at this power output.

3.10.4 Engine accessories. The radiator, radiator winterfront, cranking motor, low-temperature starting aid, air intake, and exhaust system for the plow engine shall conform to the requirements for corresponding items for the carrier engine.

3.10.5 Clutch. The rotary snowplow drive train shall include a snap-over-center or air-actuated disk-type, heavy-duty friction clutch. The clutch control shll be accessible to the driver and the plow operator. The clutch torque rating shall be not less than 130 percent of the maximum engine torque.

3.10.6 Rotary plow head. The cutting head of the rotary snowplow shall have a minimum cutting width of 8-1/2 feet and a minimum cutting height of 4 feet. Under calm wind conditions, the head shall discharge snow weighing 20 to 40 pounds per cubic foot to either side of the centerline of the vehicle a minimum distance of 20 feet out of a 14-foot-deep cut (with chute), and to a minimum distance of 125 feet out of a 4-foot deep cut (without chute). The casting and loading chutes shall be hydraulically controlled from the carrier cab. The fans, cutters, feeders, augers, blowers, and impellers, as applicable, shall be provided with individual or master shear pins that will protect the operating machinery from damage caused by foreign objects. Shear pins shall not break when plowing snow as specified herein. Shear pins shall be replaceable by personnel using common handtools while hampered with bulky clothing. Replacement shall be accomplished without removal of components, coverplates, or shrouding. Cutters, when furnished, shall be fabricated from steel of sufficient strength and heat treatment to prevent bending under capacity operations or breaking from brittleness in extremely low temperatures. With the rotary head in the transport position, the ground clearance under the plow head shall be not less than 9 inches at the lowest point. The rotary head shall be capable of being held or locked in the transport position by mechanical means, without hydraulic support. The plow mounting shall enable the snowplow to negotiate uneven surfaces and obstacles during forward and reverse travel without damage to the equipment. The means of ground support of the plow head (such as shoes, wear plates, or casters) shall be replaceable by operating personnel in the field. The loading chute shall be located on the side of the plow opposite the driver's position.

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3.10.7 Rotary hydraulic system. The system shall be capable of positioning the hydraulic-actuated equipment in any selected position within the required limits of travel and shall be of such capacity that two controls which may be operated simultaneously shall function without reduction in response time from the response time for single operation. Self-sealing and labeled, quick-disconnect fittings equipped with captive-chained dust caps shall permit rapid removal and replacement of the plow head without the necessity of bleeding the hydraulic system or the possibility of crossing hydraulic lines. The pump shall be driven by the carrier engine. The piston rods of hydraulic rams and cylinders shall be plated with chromium. The system shall operate as required at temperatures down to minus 50°F. Hydraulic pressure at any point in the system shall not exceed 2,000 psi and the maximum oil temperatures shall not exceed 200°F. Seamless corrosion-resistant steel tubing shall be used for hydraulic lines where flexing is not required. Hydraulic hose shall be protected against damage at all points of wear. Provisions shall be made for bleeding the hydraulic system.

3.10.7.1 Hydraulic tank. The hydraulic tank shall be equipped with a filler neck with removable strainer and a captive-chained cap. A filtered air vent shall be incorporated in the filler cap.

3.10.7.2 System filter. A full-flow filter shall be furnished in the system downstream from the pump. The filter shall have a reusable element that will prevent passage of particles larger than 25 microns. Pressure drop shall be not greater than 2 psi when the element is clean. The filter case shall be accessible for cleaning and inspection. An automatic bypass valve and an indicator or other warning device shall be provided to indicate when the fluid is bypassing the filter.

3.10.8 High bank cutters. When specified (see 6.2), cutters shall be furnished to remove snow up to 8 feet deep.

3.11 Fuel system.

3.11.1 Fuel tanks. The fuel system shall include tank(s) with a capacity sufficient for not less than 8 hours operation in any operating condition specified herein. Fuel capacity shall be based on 85 percent of the manufacturer's published maximum fuel consumption rate curve or 100 percent of the manufacturer's published intermittent fuel rate curve

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at the applicable horsepower and speed. The tank(s) shall: (1) conform to the requirements of the DOT Motor Carrier Safety Regulations for side-mounted diesel fuel tanks; (2) include a balance line when multiple tanks are furnished; (3) have the engine fuel intake line located to preclude starving of the engine on grades and slopes; (4) have a cap, fillpipe, and captive-chained mesh strainer which shall permit filling the tanks at a rate of not less than 20 gallons per minute; and (5) have the fillpipe located so that the tank can be filled from a 5-gallon can conforming to MIL-C-1283.

3.11.2 Fuel lines. The fuel-line system shall provide that either the carrier engine or the snowplow engine may draw fuel from either fuel tank at the option of the operator. The fuel lines shall be fastened and grommeted to prevent chafing. Exposed fuel lines shall be located to avoid proximity to high-temperature areas which could contribute to vapor lock.

3.12 Engine housings. The carrier engine, when rear-mounted, and plow engine shall each be provided with a housing fabricated of sheet steel not less than 0.071 inch in thickness. Joints and panel overlaps shall be sealed to prevent infiltration of blowing snow. The housings shall be as specified below:

(a) Each housing shall be an integral assembly and shall be capable of being unfastened from its mounting in 1 man-hour.
(b) The side panels shall open a full 180 degrees to permit access to both sides and the top of the engine and shall be equipped with a hold-open device. The panels shall also fasten in a partially opened position allowing engine cooling for hot-weather highway travel. Piano hinges equipped with 1/4-inch diameter brass or stainless steel hinge pins shall be installed at hinge points of the panels. All latches and handles shall be designed to permit operation by personnel wearing heavy gloves. Adequate clearance for the buildup of ice and snow shall be provided between the side panels and the catwalk.

3.13 Toolbox(es). The snowplow shall be provided with one or two toolboxes made of metal not less than 0.071 inch in thickness and of adequate size to accomodate the full complement of tools and tire chains. Each toolbox shall have a hinged lid and shall be equipped with not less than two fasteners. Each toolbox shall be securely fastened to the snowplow in an accessible, protected location.

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3.14 Electromagnetic interference. The electromagnetic interference emission characteristics of the snowplow shall conform to SAE J551.

3.15 Lubrication. All surfaces requiring lubrication shall be provided with a suitable means for lubricating.

3.15.1 Service products. The snowplow shall operate as specified herein when serviced with military products and antifreeze as listed in Table II.

Table II. Service Products

Item application	Specification	Type and Grade
1. Engine and transmission oils		
(a) Heavy-duty	MIL-L-2104	10, 30
(b) Arctic	MIL-L-46167	One grade
2. Fuel, diesel	VV-F-800	DF-1, DF-2 and DF-A
3. Grease		
(a) Automotive	MIL-G-10924	One grade
(b) High temperature	MIL-G-23827	One grade
4. Hydraulic-system oil (excluding brakes)		
(a) Normal ambient	MIL-L-2104	10
(b) Low temperature	MIL-L-46167	One grade
	or	
	MIL-H-5606	One grade
5. Lubricating oils		
(a) Universal	MIL-L-2105	80, 90
(b) Arctic	MIL-L-10324	One grade
(c) Instrument	MIL-L-6085	One grade
6. Antifreeze		
(a) Ethylene glycol	O-A-548	I or II
(b) Arctic antifreeze	MIL-A-11755	One grade

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3.15.2 Lubrication fittings. Lubrication fittings shall conform to SAE J534, and shall be located in a protected position. Fittings shall be accessible to a grease gun, with a 10-inch flexible extension. Fittings shall be accessible without removal or adjustment of accessories or parts. Panels and plates equipped with hand-operated, quick-disconnect fasteners may be removed to provide accessibility.

3.15.3 Pressure-release device. Where the use of pressure lubricating equipment could damage grease seals or other parts, an automatic pressure-release device shall be provided.

3.15.4 Inclosures with integral reservoir of lubricant. Inclosures such as gearcases and transmission housings which contain a reservoir of lubricants shall afford clearance for arctic-clothed personnel to check the level of the lubricant. Each inclosure shall be equipped with a magnetic drain plug and means for filling the inclosure with lubricant from a 1-quart, hinged-spout oilcan. The drain plug shall be located so that removal of the plug will result in complete drainage of the lubricant from the inclosure. Drainage shall be to the ground when the equipment is in its normal position. Drain plugs, filling means, and the lubricant-level checking devices shall be accessible without removal or adjustment of components. Panels and plates equipped with hand-operated quick-disconnect fasteners may be removed to provide accessibility.

3.15.5 Delivery lubrication service. Unless otherwise preserved for packaging (see 5.2), all components shall be filled to the operating level with winter grades of military oils designated for use in the temperature range of 0°F to 32°F. Grease lubrication, including lubrication of sealed bearings, shall be with grease conforming to MIL-G-10924 or MIL-G-23827, as applicable. Snowplows shall be tagged in a conspicuous place in the operator's compartment indicating the temperature range and grades of oils and greases used.

3.16 Fungus and moisture resistance. The electrical circuitry, including all components and connections except for the following, shall be protected from the effects of moisture and fungus growth by an overall treatment with varnish conforming to MIL-V-173, composition as specified in 3.16.1, or a varnish having a 1 percent copper 8-quinolinolate content (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

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(b) Components or circuit elements whose funtions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or functioning of the varnish. Insulating materials shall either be inherently resistant to fungus and moisture or shall be protected from the effects of moisture and fungus.

3.16.1 Composition. Composition I shall be used except when Composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When Composition II is used, the contractor shall provide evidence to the Government that its use is required, and shall certify that the Composition II material complies with Rule 66 - Air Pollution Control District, County of Los Angeles, CA.

3.17 Fire extinguisher. A 5-pound CO_2 fire extinguisher shall be provided and mounted in the carrier cab. The extinguisher shall be mounted so as to be accessible to the operators.

3.18 Plow head transport. Lifting and tiedown points shall be provided on the rotary plow head. The provisions shall permit lifting and securing the assembly during transport operations without damage when separated from the carrier. A shipping instruction plate conforming to 3.20 shall be attached to the plow head.

3.19 Identification marking. The snowplow shall be identified in accordance with MIL-STD-130. The marking shall be applied to the carrier on identification plates conforming to MIL-P-514, Type I, photographic plate. A tranportation-data plate conforming to MIL-P-514, Type III, photographic plate, shall be furnished and shall indicate the location of lifting and tiedown points and the longitudinal and vertical locations of the center of gravity.

3.20 Instruction plates. The snowplow shall be equipped with instruction plates describing any special or important procedure to be followed in operating and servicing, and in preparation for shipment. Plates shall conform to MIL-P-514, Type III, photographic plate.

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3.21 Treatment and painting. The portions of the snowplow normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, except that commercial cleaning, treating, and prime coating will be acceptable provided they comply with the following:

(a) The commercial paint and application process is equal to or better than the cleaning, pretreatment, and priming required by MIL-T-704.
(b) The commercial paint supplied is compatible with the finish coat as required by MIL-T-704. Unless otherwise specified (see 6.2), the finish color shall be No. 24087 of FED. STD. No. 595.
(c) The finished paint thickness will be equal to or greater than the total thickness required by MIL-T-704 for prime and finish coats. Parts subject to temperatures in excess of 450°F shall be painted in accordance with the contractor's standard practice.
(d) Crankcase sealer - The application of a crankcase sealer as specified in MIL-T-704 on the interior surfaces need not be appliced on the cast iron housings of transmission and torque converters provided the interior surfaces are void of casting defects such as blowholes, excessive porosity, cold shuts, hard spots, shrinks, cracks, or other injurious defects.

3.21.1 Undercoating. The wheel wells, metal fenders, and other sheet metal directly exposed to road splash shall be coated with commercial sandless undercoating material.

3.22 Stenciling. The GVWR of the snowplow and the longitudinal location of the center of gravity shall be stenciled on both sides of the snowplow. The weight of the plow head shall be stenciled on both sides of the plow head. The prescribed tire pressure shall be stenciled on the carrier near each wheel in letters not more than 1 inch high. Unless otherwise specified herein, stenciling shall conform to MIL-T-704. "Diesel fuel" shall be stenciled in letters not more than 1 inch high near the fuel tanks.

3.23 Class A, nonwinterized. The nonwinterized snowplow shall conform to 3.1 through 3.22.

3.24 Class B, winterized. The winterized snowplow shall conform to all requirements applicable to the nonwinterized unit. In addition, provisions shall be furnished to permit starting and operation at temperature

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and elevation conditions specified in 3.4.2. Provisions shall include, but not necessarily be limited to the following:

(a) Plug in type electrical heaters with thermostats for engine coolant and engine oil sump of both engines and for the batteries. The heater wattage shall conform to Table III. The electrical connections for the heaters shall be combined, where practical, to minimize the number of external connections. Each external connector shall have a maximum current draw of 30 amps with a 115-volt supply. All the external electrical connections shall be located in the same general area of the snowplow to allow easy access.
(b) Increased capacity of cab heater-defroster installation to meet the performance required.
(c) Additional thickness and area of cab insulation.
(d) Additional detailed winterization requirements shall be as specified (see 6.2).

TABLE III

	Minimum Wattage Rating Engine Coolant Heater	Minimum Wattage Rating Engine Oil Sump Heater
Plow Engine	4000	600
Carrier Engine	3000	300

3.25 Workmanship. The snowplow shall be free from incomplete welds, rust, cracks, and other defects that could impair the operation or serviceability of the unit. The quality of workmanship on the snowplow shall be as good as, or better than, the quality the contractor delivers with his commercial production of similar items.

3.25.1 Steel fabrication. Steel used in the fabrication of the equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and clipping shall be done neatly and accurately.

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3.25.2 Welders and welding.

3.25.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

3.25.2.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. (Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.) Parent materials, weld filler metals and fabrication techniques shall be as required to enable the snowplow to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. The welding process used in fabrication of the snowplow shall be at the option of the contractor.

3.25.2.3 Welding practices.

(a) Preheat of materials being welded and maximum interpass temperature during welding shall be in accordance with the contractors recommendations.
(b) Tack welds shall be subject to the same quality requirements as final welds.
(c) Work shall be positioned for flat welding whenever practicable.

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(d) Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and snowplow.

3.25.2.4 Weld quality.

(a) The contractor shall be responsible for determining the inspections needed to insure that, when the welded parts are assembled together to make the snowplow, the snowplow shall conform to the inspection requirements specified in Section 4.
(b) All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to either the appearance or strength of the weldment.
(c) Undercut in weldments shall be held to a minimum and shall not be more than 1/32 inch deep.

3.26 Technical information package. The contractor shall complete and submit a technical information package within the time frame specified (see 6.2). The information shall depict the complete snowplow and its components exactly as it will be furnished.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.5).

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4.3 Preproduction inspection. Prior to examination and test of the snowplow, the following shall be performed:

(a) After installation of the engines in the carrier, service the snowplow with military oils, greases, and fuel specified in 3.15 and designated for use in the ambient temperatures at which the tests will be run.
(b) Limber up the snowplow as recommended by the contractor.

4.3.1 Examination.

4.3.1.1 Pretest examination. Prior to testing, the completed technical information package shall be reviewed. Any discrepancies between the completed technical information package, the specification, and the preproduction model shall be cause for rejection. Prior to testing, the snowplow shall be examined for the defects marked "X" in Column 1 of Table IV. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Posttest examination. Upon completion of all tests, the snowplow shall be examined in accordance with 4.5.2.22. Failure of this examination shall be cause for rejection.

4.3.2 Preproduction tests. Upon successful completion of the examination specified in 4.3.1, the snowplow shall be subject to tests indicated in Column 2 of Table V. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Individual examination. After successful completion of the test specification in 4.4.2, each snowplow shall be examined for the defects marked "X" in Column 2 of Table IV. Presence of one or more defects shall be cause for rejection.

4.4.2 Individual tests. Each snowplow shall be tested as specified in 4.5.2.21. Failure of the test shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. The examination shall be in accordance with Table IV.

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Table IV. Examination Schedule

Pre-prod.	Indiv.	Defect	Requirement paragraph(s)
1	2	3	4
X	-	101. Dimensions or weights not as specified.	3.1, 3.8, 3.9.14
X	X	102. Assembly not complete or not correct.	Section 3
X	X	103. Material not as specified.	3.3, Section 3
X	X	104. Safety devices or guards missing or not as specified.	3.5, 3.10.2
X	X	105. Lubrication devices not accessible	3.15.2
X	-	106. Battery box not as specified.	3.9.18.2
X	-	107. Drive train not as specified.	3.9.10
X	-	108. Final drive not as specified.	3.9.10.3
X	-	109. Chassis not as specified.	3.11.3, 3.9.11
X	-	110. Radiator not as specified.	3.9.8, 3.10.4
X	-	111. Electrical system not as specified.	3.9.18
X	X	112. Lighting not as specified.	3.9.18.1
X	-	113. Cab not as specified.	3.9.20
X	X	114. Identification or other marking or operating and safety instructions missing.	3.19, 3.20
X	X	115. Treatment and painting not as specified.	3.21
X	-	116. Material of incorrect thickness.	3.3, 3.25
X	X	117. Workmanship not as specified.	3.25
X	X	118. Center of gravity markings omitted.	3.22
X	-	119. Technical information package not completed.	3.26
X	-	120. Ease of maintenance provisions not as specified.	3.6

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4.5.2 Tests. Tests shall be in accordance with Table V.

Table V. Test Schedule

Item No.	Pre-prod.	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5
1	X	Draw bar pull.	4.5.2.2	3.9.12
2	X	Service brakes.	4.5.2.3	3.9.17
3	X	Parking brake.	4.5.2.4	3.9.17.2
4	X	Vehicle clearance circle.	4.5.2.5	3.8
5	X	Rocking test.	4.5.2.6	3.9.10.2
6	X	Torque converter stall.	4.5.2.7	3.9.10.1
7	X	Emergency steering.	4.5.2.8	3.9.16, 3.9.10.2
8	X	Steering.	4.5.2.9	3.9.16
9	X	Road travel.	4.5.2.10	3.9.10
10	X	Travel speed.	4.5.2.11	3.9.12
11	X	Sound level.	4.5.2.12	3.5.2.1
12	X	Low temperature.	4.5.2.13	3.4
13	X	Defroster test.	4.5.2.14	3.9.21
14	X	Shear pin.	4.5.2.15	3.10.6
15	X	Performance.	4.5.2.16	Section 3
16	X	Casting ability production.	4.5.2.17	3.7.1, 3.7.2, 3.10.6
17	X	Rated capacity.	4.5.2.18	3.7.1
18	X	Wheel torque transfer.	4.5.2.19	3.9.10.3
19	X	Electromagnetic interference.	4.5.2.20	3.14
20	X	Functional.	4.5.2.21	Section 3
21	X	Posttest examination.	4.5.2.22	Section 3

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4.5.2.1 Test conditions.

(a) Test apparatus shall be accurate within the limits specified in SAE J872.
(b) Test methods referenced herein are those contained in the Appendix, Part I.
(c) All lights shall be turned on and left on during all tests.
(d) The snowplow shall operate as specified herein without maintenance other than the contractor's recommended scheduled maintenance as established by a maintenance schedule submitted by the contractor prior to test.
(e) Throughout the tests, evaluate safety and human factors considerations for compliance with 3.5.

4.5.2.2 Drawbar pull. Test the snowplow with Test Method No. 1. Nonconformance to 3.9.12 shall constitute failure of this test.

4.5.2.3 Service and emergency brakes. Test the snowplow in accordance with Test Method No. 2. Nonconformance to DOT FMVSS, DOT MCSR, or the California Vehicle Code as specified in 3.9.17, inability of the brakes to bring the snowplow to a complete stop safely or loss of contact of any wheel with the ground during the braking stops shall constitute failure of this test.

4.5.2.4 Parking brake. Test the snowplow in accordance with Test Method No. 3. A brake lever pull or push in excess of 90 pounds or failure of the brake to hold when subjected to the required pull (see 3.9.17.2) shall constitute failure of this test.

4.5.2.5 Vehicle clearance circle. Test the snowplow in accordance with Test Method No. 4. A turning clearance greater than that specified in 3.8 shall constitute failure of this test.

4.5.2.6 Rocking test. Test the snowplow in accordance with Test Method No. 5. Nonconformance to 3.9.10.2 shall constitute failure of this test.

4.5.2.7 Torque converter stall. Test the snowplow in accordance with Test Method No. 6. Nonconformance to 3.9.10.1 shall constitute failure of this test.

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4.5.2.8 Emergency steering. Test the snowplow in accordance with Test Method No. 7. Nonconformance to 3.9.16 or 3.9.10.2 shall constitute failure of this test.

4.5.2.9 Steering test. Test the snowplow in accordance with Test Method No. 8. Nonconformance to 3.9.16 shall constitute failure of this test.

4.5.2.10 Road travel. Test the snowplow in accordance with Test Method No. 9. Nonconformance to 3.9.10 shall constitute failure of this test.

4.5.2.11 Travel speed. Test the snowplow in accordance with Test Method No. 10. Nonconformance to the speed ranges specified in 3.9.12 shall constitute failure of this test. This test (or parts of this test) may be combined with the test in 4.5.2.10.

4.5.2.12 Sound level. Test the snowplow in accordance with Test Method No. 11. Nonconformance to 3.5.2.1 shall constitute failure of this test.

4.5.2.13 Low temperature. When Class A or Class B snowplow is required, test the snowplow in accordance with Test Method No. 12. Nonconformance to 3.4 shall constitute failure of this test. (This test may be combined with test in 4.5.2.14).

4.5.2.14 Defroster test. Test the snowplow in accordance with Test Method No. 13. Nonconformance to 3.9.21 shall constitute failure of this test. (This test may be combined with the test in 4.5.2.13).

4.5.2.15 Shear pins. Test the snowplow in accordance with Test Method No. 14. Nonconformance to 3.10.6 shall constitute failure of this test.

4.5.2.16 Performance. Test the snowplow in accordance with Test Method No. 15. Inability of the snowplow to operate as specified or evidence of breakage, permanent deformation, or failure of components and parts, except for shear pins if an obstruction is encountered, shall constitute failure of this test. This test may be combined with the tests in 4.5.2.17 and 4.5.2.18.

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4.5.2.17 Casting ability. Test the snowplow in accordance with Test Method No. 16. Inability to meet the requirements in 3.7.1, 3.7.2, and 3.10.6 shall constitute failure of this test. This test may be combined with tests in 4.5.2.16 and 4.5.2.18.

4.5.2.18 Rated production capacity. Test the snowplow in accordance with Test Method No. 17. Production rates of less than 25 tons per minute shall constitute failure of this test. This test may be combined with the tests in 4.5.2.15 and 4.5.2.16.

4.5.2.19 Wheel torque transfer. Test the snowplow in accordance with Test Method No. 18. Nonconformance to 3.9.10.3 shall constitute failure of this test.

4.5.2.20 Electromagnetic interference. The snowplow shall be tested to determine compliance with 3.14. Failure of the snowplow to meet the specified requirement shall constitute failure of this test.

4.5.2.21 Functional test. Each snowplow shall be tested in accordance with Test Method No. 19. Nonconformance to 3.9.16, 3.9.17, malfunction of any control, or instrument, or evidence of need for adjustment shall constitute failure of this test.

4.5.2.22 Posttest examination. The snowplow shall be examined visually upon completion of all required tests. Any of the following shall constitute failure:

(1) Excessive wear.
(2) Evidence of permanent deformation.
(3) Lack of lubrication.
(4) Breakage.
(5) Malfunction of components.

4.6 Inspection comparison. The Government may select snowplows at any time during the contract production period and subject the snowplows to the examination specified in 4.5.1 and Column 2 of Table IV, and to the tests specified in Column 2 of Table IV, to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will

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not include the previously inspected preproduction model and initial production snowplows. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison snowplow to meet any requirement specified herein during and as a result of the examination and test specified in 4.6 shall be cause for rejection of the inspection comparison snowplow(s) and shall be cause for refusal by the Government to continue acceptance of production snowplows until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at not cost to the Government on snowplows previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all snowplows accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging. The preservation and packing shall be examined to determine compliance with the Quality Assurance Provisions of MIL-STD-281. The marking shall be examined to determine compliance with the referenced document. Any marking that is missing, illegible, incorrect, or incomplete shall be cause for rejection.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

5.1.1 Level A. Each complete snowplow shall be preserved in accordance with the Level A (mobile) requirements of MIL-STD-281, with the following exceptions and additional requirements. Components not specifically specified shall be preserved as specified for similar type components. Materials and methods required for preservation specified herein shall conform to the applicable specifications listed in MIL-STD-281.

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5.1.1.1 Preservative. Preservative oil specified in MIL-STD-281 as P-10, Grade 1 or 2, shall conform to P-10, Type I, Grade 10 or 30 as applicable.

5.1.1.2 Engines. The carrier engine and rotary plow engine shall both be preserved in accordance with the Level A requirements of MIL-E-10062, Type I.

5.1.1.3 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.1.1.4 Maintenance tools. Maintenance tools shall be preserved in accordance with the Level A requirements of PPP-P-40.

5.1.1.5 Batteries. Batteries shall be secured in place in the battery carrier as specified in MIL-STD-281.

5.1.1.6 Hydraulic system (except steering system). The hydraulic fluid supply tank shall be filled with the hydraulic fluid recommended for operation. The pistons shall be retracted as far as practicable into the cylinder and secured. When the pistons cannot be fully retracted, exposed portions of the piston rods (ramshafts) shall be coated with Type P-6 or P-11 preservative and wrapped or covered with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2; extending the wrap approximately 2 inches on the ram cylinder, if possible. The wrap shall be secured in place with tape conforming to PPP-T-60, Type IV. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with Type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position. Any unpainted exterior metal surfaces of the system requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative . A tag conforming to UU-T-81, Type A shall be attached to the control lever indicating:

> "The hydraulic supply tank is filled with the fluid required for operation."

The printing thereon shall be with waterproof ink.

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5.1.2 Level B. Each complete snowplow shall be preserved in accordance with the Level B (mobile) requirements of MIL-STD-281, with the exceptions and additional requirements specified in 5.1.1 for Level A.

5.1.3 Commercial. Each complete snowplow shall be preserved in accordance with FED. STD. No. 356. In addition, the crankcase shall contain the required amount of lubricating oil and the cooling system shall be filled with a fresh solution of 50 percent water and 50 percent ethylene glycol type antifreeze.

5.2 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A. The carrier and all associated components, preserved as specified in 5.1, shall not require additional packing.

5.2.2 Commercial. Each complete snowplow shall be packed in accordance with FED. STD. No. 356.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with FED. STD. No. 356.

6. NOTES

6.1 Intended use. The snowplow is intended for use in cleaning snow from highways, streets, hardstands, airport runways, taxiways, and parking aprons; and for casting the snow or loading it into trucks for disposal.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Time frame required for submission of preproduction model (see 3.2).

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(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(e) When cutters are required (see 3.10.8).
(f) When finish paint color other than 24087 is required (see 3.21(b)).
(g) Additional winterization requirements for Class B snowplows (see 3.24).
(h) Degree of preservation and of packing required (see 5.1 and 5.2).
(i) Time frame for completing and submitting the technical information package (see 3.26).

6.3 Preproduction model. Any changes or deviations of production snowplows from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish snowplows conforming to this specification.

6.4 Cold chamber tests. The contracting officer may request that the cost of the preproduction model test and the cost of those tests requiring a cold chamber (Test Method No. 12 and Test Method No. 13) be priced as separate items in the bid or proposal, for evaluation purposes.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional material, illustrated parts list, and contractor's maintenance and operation manual to be furnished with each snowplow.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions on shipment of snowplows. A suggested paragraph is as follows:

(Shipment of snowplows shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

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Custodian:
Army - ME

Review activity:
Army - AT

Preparing activity:
Army - ME

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APPENDIX

FOR

SNOWPLOW, ROTARY

TRUCK-MOUNTED, DED

PART I. TEST METHODS

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TEST METHOD NO. 1

DRAWBAR PULL

1. TEST COURSE

The test course shall be clean dry paved surface with not more than 1 percent average longitudinal grade. The crown slope shall not exceed 1.5 percent, measured from center to outer edge. The width of the straight portions shall be wider than the vehicle and long enough to meet the requirements of the test method.

2. TEST APPARATUS

(a) General apparatus: (Alternate approved equipment may be used).

(1) Barometer.
(2) Gage, tire pressure.
(3) Load cell, electronic, w/amplifier and recorder (use lock-out device, when possible, to prevent overload of instruments during turns and accelerations), or other approved drawbar measuring equipment.
(4) Odometer.
(5) Scale, platform.
(6) Tachometer.
(7) Tape, measuring, or surveyor's chain.
(8) Thermometer.
(9) Watch, stop.
(10) Electronic recorder to obtain speed and elapsed time.
(11) Robert Bosch sampling pump - EFAW 65.
(12) Robert Bosch evaluating instrument - EFAW 68 (the Clayton opacity meter may be used).

(b) Special apparatus: (A towed load is required with a drag force adjustable as necessary to control).

(1) The engine speed of the towing vehicle constant within plus or minus 3 percent of a selected speed, or
(2) The drag force constant within plus or minus 2-1/2 percent of the selected load.

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3. TEST PROCEDURES

(a) Connect a load cell in series with a cable connected between the pintle hook and the towed vehicle or vehicles used to apply the load. The test cable shall be connected to the towed load so as to be parallel to the ground.

(b) With the engine operating at full throttle, perform a series of pulls in each forward gear at varying loads. A minimum of one pull in each direction of the test course for each load shall be required. The number of pulls shall be sufficient to determine the maximum horsepower in each gear and to plot pull versus ground speed (see Table I).

(c) Determine the travel efficiency during each pull by comparing the number of revolutions of the same wheel over the test course when the snowplow is not under load.

(d) Record for each pull:

(1) Load, pounds - reserve tractive force.
(2) Ground speed, mph.
(3) Travel efficiency.
(4) Engine speed, rpm.

(e) Maintain the snowplow in fully serviced condition throughout the test. Inflate tires in accordance with contractor's specifications. Take smoke density measurements at or near maximum power point in first and second gear.

(f) Compute the maximum power point in each gear by the following:

$$\text{Drawbar horsepower} = \frac{\text{Drawbar pull (lbs) X Speed (mph)}}{375}$$

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TEST METHOD NO. 2

SERVICE AND EMERGENCY BRAKES

1. TEST COURSE

The test course shall be a uniform improved surface allowing no penetration, with a grade not exceeding plus or minus 1 percent longitudinally and shall be sufficiently long to permit the snowplow to attain DOT FMVSS speeds.

2. TEST APPARATUS

(a) Tape, measuring (100-foot).
(b) Chalk.
(c) Detonator, 2-barrel, brake apparatus.
(d) Blanks, .22 cal., yellow powder.
(e) Towed fifth wheel.

3. TEST PROCEDURE

(a) Connect the detonator as follows:

(1) First barrel - actuated by a switch directly connected to the brake pedal so that it is actuated during the first 1/4 inch of pedal travel.
(2) Second barrel - actuated by a switch operated manually after the snowplow has come to a complete stop.

(b) Conduct tests in accordance with DOT FMVSS, DOT MCSR, and the California Vehicle Code without plow head. Repeat similar tests with the plow head installed.

Note: A brake tester apparatus approved by the Government may be used in place of the detonator and switches.

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TEST METHOD NO. 3

PARKING BRAKE

1. EQUIPMENT

Prime mover, or other means of developing the required pull; 100-pound-capacity spring scale; tension load cell and appropriate instrumentation.

2. TEST PROCEDURE

(a) Stop the snowplow, with plow head raised, on a level, concrete surface. Apply the parking brake and place the transmission in neutral. Exert a force equal to 50 percent of the snowplow weight at the front of the snowplow in a direction parallel to the longitudinal axis of the snowplow and away from the snowplow. Repeat the test with the force being exerted at the rear of the snowplow.

(b) As an alternate method, the holding ability of the parking brake can be determined on a longitudinal slope of 60 percent if applied in both the descending and ascending attitude. On a 60 percent slope the required holding force of the parking brake is comparable to the required 50 percent of the vehicle weight.

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TEST METHOD NO. 4

VEHICLE CLEARANCE CIRCLE

1. TEST EQUIPMENT

(a) 100-foot measuring tape.
(b) Plumb line or rod attached to the snowplow in vertical position.
(c) 6-foot straight edge.
(d) Markers and chalk.
(e) Optionally, distance measuring wheel (see 3(g) below).

2. TEST COURSE

The course shall be dry, level concrete or firm ground large enough to accommodate the carrier when negotiating a turn of not less than 270 degrees.

3. TEST PROCEDURES

(a) Adjust tire pressures to recommended operating pressure.
(b) Drive the carrier slowly in a sharp curve and determine the part of the carrier which extends farthest from the center of the turning circle. Attach the plumb bob or vertical rod at that point.
(c) Operate the carrier to make a turn to the right of not less than 270 degrees while traveling forward in low gear with the engine at idle speed.
(d) During step (c) mark on the course the path described by that part of the carrier which extends farthest from the center of the circle.
(e) Measure and record the diameter of the marked path in three directions not less than 20 degrees apart. Take the average of the three measurements, record this average measurement.
(f) Repeat (c), (d), and (e) above, except the 270-degree turn shall be made to the left.
(g) As an alternative to (a) through (e), mount a measuring trailing wheel at that part of the snowplow which extends the farthest from the center of the circle. Record the distance to describe a 360 degree arc. This is the circumference. Compute the diameter of the circle.

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TEST METHOD NO. 5

ROCKING TEST

1. TEST COURSE

The test course shall be a dry, level surface of sufficient length for the snowplow to perform the following test.

2. TEST APPARATUS

(a) Stop watch.

3. TEST PROCEDURE

(a) With the transfer case in highest gear, perform not less than six of the following cycles in not more than 120 seconds as follows: Starting with the main transmission in low gear forward, apply wide-open-throttle for 5 seconds, release throttle for 4 seconds, shift transmission into reverse, apply wide-open-throttle for 5 seconds, release throttle for 4 seconds and shift into forward.

(b) With the transfer case in highest range, shift into low gear forward and apply wide-open-throttle for a distance of at least 20 feet, shift into reverse without letting up on the throttle, travel a distance of at least 20 feet backward, and shift into forward without letting up on the throttle.

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TEST METHOD NO. 6

TORQUE CONVERTER STALL

1. TEST COURSE

The test course shall be concrete or other hard surface. The width shall be wider than the vehicle and long enough to meet the requirements of the test method.

2. TEST APPARATUS

(a) General apparatus: (Alternate approved equipment may be used.)

(1) Stop watch.
(2) Thermocouples and appropriate instrumentation.

3. TEST PROCEDURE

The ambient temperature shall be plus 30°F or above.

(a) Bring the torque converter oil temperature to at least plus 130°F.
(b) With the carrier engine operating at wide-open-throttle, stall the torque converter for not less than 30 seconds.
(c) Operate the snowplow in a forward gear not to exceed 5 mph until the torque converter oil temperature is stabilized between 130°F and 180°F.
(d) Repeat (b) and (c) above three more times, except the stall shall be 15 seconds.
(e) Record:

(1) Oil temperature at start of test.
(2) Ambient temperature.
(3) Oil temperature at end of stall.
(4) Oil temperature at stabilized operating temperature.

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TEST METHOD NO. 7

EMERGENCY STEERING

1. TEST COURSE

The test course shall be a dry, level surface of sufficient length for the snowplow to attain a speed of not less than 25 mph. The course shall also include a curve with a maximum centerline radius of 100 feet in accordance with Figure 1.

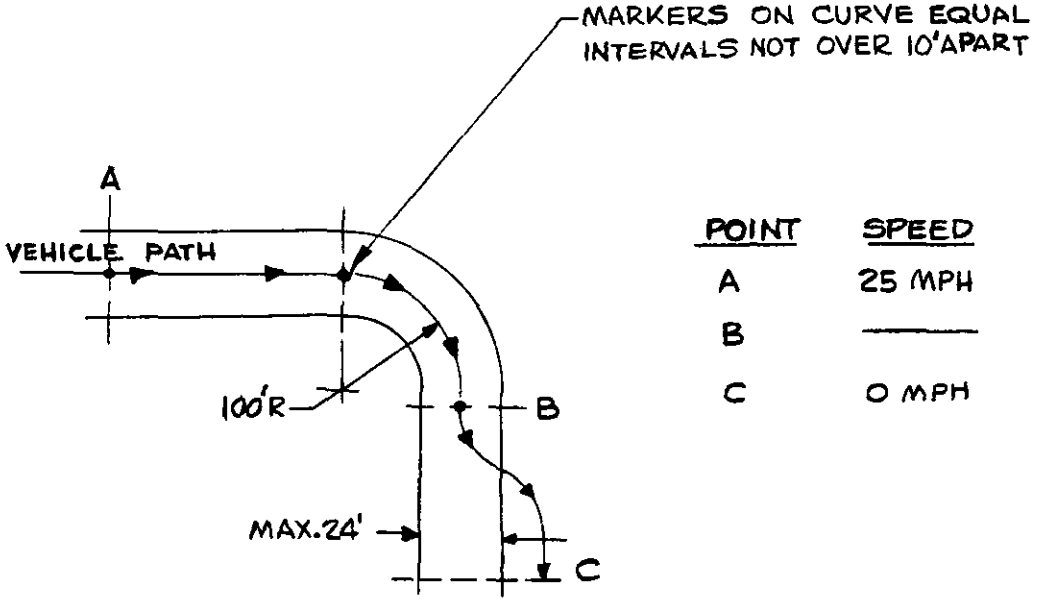
2. TEST APPARATUS

(a) Tape, measuring (100-foot).
(b) Markers (20 each of sufficient length to extend 36 inches above ground level), equally spaced along curve centerline.
(c) Speed indicator.
(d) Towing vehicle and tow cable.

3. TEST PROCEDURE

(a) Drive the snowplow a sufficient distance to attain a speed of not less than 25 mph. When the snowplow reaches point A (Figure 1) traveling 25 mph, shut off the engine, shift the transmission into neutral. Slow the snowplow by use of the service brakes if necessary. At point B, steer to the outside of the road away from the curve and pull off on the shoulder of the road. Stop at point C. The driver shall maintain complete control of the snowplow at all times during the test. All markers on the 100-foot radius curve shall be contacted by the snowplow while negotiating the curve.
(b) Attach the towing vehicle to the snowplow using the tow cable. Do not start the carrier engine; place the transmission in neutral. Tow the snowplow through the course (Figure 1) at a speed of 15 mph, steering as in (a) above. Stop at point C by using the carrier brake system.

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FIGURE 1. TEST COURSE FOR EMERGENCY STEERING TEST.

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TEST METHOD NO. 8

STEERING

1. TEST COURSE

The test course shall be of a uniform, improved surface of sufficient size to accommodate the snowplow to perform the following test.

2. TEST EQUIPMENT

(a) Spring scale.
(b) Tire pressure gage.

3. TEST PROCEDURES

The snowplow shall be serviced and checked for recommended tire pressure.

(a) With hydraulic pressure and engine running, measure and record:

(1) Effort required to turn the steering wheel from lock-to-lock position under static conditions.
(2) Effort required to turn the steering wheel when vehicle is moving.

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TEST METHOD NO. 9

ROAD TRAVEL

1. TEST COURSE

Paved or improved surface test course with the length not exceeding 20 miles.

2. TEST APPARATUS

(a) Thermocouples and appropriate instrumentation.
(b) Timing devices.

3. TEST PROCEDURE

(a) Install thermocouples to measure liquid temperatures;

(1) In the oil discharge line from the converter to the cooler.
(2) In the converter sump.
(3) In the engine oil gallery.
(4) In the hydraulic tank.
(5) In the transfer case sump.
(6) In the transmission sump.
(7) In the front and rear differentials.
(8) In the engine coolant discharge line.

(b) Drive the snowplow, at maximum speed compatible with safety, in the forward direction for not less than 100 miles or until the temperatures stabilize.
(c) During the run, record continuously or at intervals not greater than 15 minutes, the temperatures measured at the locations specified in (a) above.

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TEST METHOD NO. 10

TRAVEL SPEED

1. TEST COURSE

The test course shall be a dry, level surface of sufficient length to attain maximum speed prior to entering measured distance.

2. TEST APPARATUS

(a) Gage, tire pressure.
(b) Tape, measuring, surveyors.
(c) Watch, stop.

3. TEST PROCEDURE

(a) Maintain the tires at pressure recommended by the contractor.
(b) Operate the carrier engine at full throttle.
(c) Drive the snowplow a sufficient distance to attain maximum speed prior to entering the measured test course.
(d) Conduct a total of six runs, three in each direction in each gear and range.
(e) Travel speed in each gear shall be the average of six runs.

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TEST METHOD NO. 11

SOUND LEVEL

1. TEST FACILITY

In accordance with SAE J919 and SAE J88.

2. TEST APPARATUS

In accordance with SAE J919 and SAE J88.

3. TEST PROCEDURE

(a) Measure the sound level in accordance with SAE J919. The plow engine shall be run at full throttle with no load during this test.
(b) Measure the sound level in accordance with SAE J88. The plow engine shall be run at full throttle with no load during this test.

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TEST METHOD NO. 12

LOW TEMPERATURE START TEST

1. TEST FACILITY

The test facility shall be a test chamber of sufficient size and capacity to contain the snowplow and to provide an ambient temperature of minus 25°F or minus 50°F, as required.

2. TEST APPARATUS

(a) Thermocouples and appropriate instrumentation.
(b) Timing device.
(c) Appropriate cold weather clothing.

3. TEST PROCEDURES

Service the snowplow with appropriate antifreeze, lubricant, fuel and hydraulic oil as specified. Install thermocouples at all points required to read the temperatures specified. The chamber exhaust system shall remove all engine exhaust fumes from the chamber during the test. Subject the snowplow to a temperature of minus 25°F or minus 50°F, as required, until stabilized.

(a) Start both the plow and carrier engines. For minus 25° F starts, the engines shall be started without engine or battery preheating. For minus 25° F starts, the engines shall be cranked from the cold soaked batteries of the vehicle only. Substitute batteries or other electrical cranking energy shall not be used. Ether or glow plugs may be used, when installed and furnished with the snowplow.
(b) Secure smooth no-load running without unnatural or continued control manipulation within 15 minutes after starting of each engine.
(c) Engage power train to snowplow and actuate the plow controls.
(d) Maintain specified engine operating temperatures.
(e) Check test instruments and gages to assure proper registering.
(f) Check accessibility to controls, adjustable devices, and valves when operators are hampered by bulky, arctic clothing including heavy gloves and boots.

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(g) Cause torque converter to stall momentarily for between 2 and 3 seconds.
(h) Record temperatures at the following locations:

(1) Engine cooling temperature in engine cooling jacket discharge pipe and cooling inlet to the engine.
(2) Engine lubricating oil temperature in oil gallery.
(3) Torque converter oil temperature at the outlet from the converter to the cooler.
(4) Hydraulic oil temperature at the midpoint of the reservoir.
(5) Ambient temperature at a horizontal angle of 45 degrees from the sides and end of the snowplow and not more than 3 feet from each corner of the snowplow at engine crankshaft level.

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TEST METHOD NO. 13

WINDSHIELD AND WINDOW DEFROSTER TEST

1. TEST FACILITY

The test facility shall be a test chamber of sufficient size and capacity to contain the snowplow and to provide temperatures to minus 25°F and minus 50°F, as required.

2. TEST APPARATUS

(a) Thermocouples and appropriate instrumentation.
(b) Timing device.
(c) Appropriate cold weather clothing.
(d) Spray gun with air supply.
(e) Measuring container.

3. TEST PROCEDURES

The snowplow shall be serviced as in Test Method No. 12. The test shall be performed as follows:

(a) The temperature of the vehicle shall be stabilized at not higher than minus 25°F or minus 50°F, as required.
(b) One quart of water for each 500 square inches of glass area shall be uniformly sprayed on the outside surface of the glass from a paint spray gun operated at 60 psi.
(c) The vehicle shall be further cold-soaked for not less than 15 minutes.
(d) The defroster system, including circulating fans in the cab, shall be started.
(e) At the end of each 15-minute period of defroster operations, the slush line shall be marked and labeled on the windshield only.
(f) After an operation time limit of 1 hour, the slush line shall have passed not less than 75 percent of each area of the windshield and of all windows. At this time, also, the area of the windshield only shall be photographed showing the defroster pattern.

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TEST METHOD NO. 14

SHEAR PINS

1. TEST EQUIPMENT

(a) Handtools.
(b) Arctic clothing.
(c) Shear pins.
(d) Block(s).

2. TEST COURSE

The course shall be a level area large enough to accommodate the snowplow.

3. TEST PROCEDURES

(a) Block the augers on sides.
(b) Start the plow engine and engage the plow drive. Observe and record results.
(c) With heavy gloves and jacket on, replace the shear pins, using only common handtools.
(d) Repeat the above tests for each set of shear pins furnished, with the corresponding plow head component restrained.

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TEST METHOD NO. 15

PERFORMANCE TEST

1. TEST COURSE

The test facility shall be an area of sufficient size with depth of 1 to 4 feet of snow of the density specified in 3.9.1.

2. TEST APPARATUS

(a) Stop watch.
(b) Measuring tape.
(c) Platform scale.
(d) Snow-sampling device approved by the contracting officer.

3. TEST PROCEDURE

Operate the snowplow for a period of 10 hours plowing snow having a depth of not less than 1 foot.

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TEST METHOD NO. 16

CASTING ABILITY

1. TEST COURSE

The test facility shall be an area of sufficient size with adequate depth (1 to 4 feet) of snow of density specified in 3.9.1.

2. TEST APPARATUS

(a) Stop watch.
(b) Measuring tape.
(c) Platform scale.
(d) Snow-sampling device approved by the contracting officer.

3. TEST PROCEDURES

(a) While the snowplow is plowing snow at the rate of not less than 25 tons per minute, both right- and left-hand casts for distances of 20 and 125 feet shall be performed each for an operating distance of at least 100 feet.
(b) Demonstrate the ability of the snowplow to spotcast in the range of 14 to 60 feet.

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TEST METHOD NO. 17

RATED PRODUCTION CAPACITY

1. TEST COURSE

The test facility shall be an area of sufficient size with adequate depth (1 foot to 4 feet) of snow of density specified in 3.9.1.

2. TEST APPARATUS

(a) Stop watch.
(b) Measuring tape.
(c) Platform scale.
(d) Snow-sampling device approved by the contracting officer.
(e) Robert Bosch EFAW 65 and EFAW 68 exhaust smoke instruments. See Test Method No. 1, 2 (a) (11) and (12).

3. TEST PROCEDURES

The snowplow shall be tested to determine the production rate of 25 tons per minute by plowing snow having a minimum depth of 1 foot for a distance of not less than 5000 feet to determine compliance with 3.9.1. The production rate shall be not less than 25 tons per minute as computed using the following formula:

$$\text{Production rate (tons per minute)} = \frac{LWDd}{(2{,}000)\ (t)}$$

L = Length of test lane, feet.
W = Width of path, feet.
D = Average depth of snow, feet.
d = Snow density, pounds per cubic foot, average for full depth.
t = Time to traverse test lane, minutes.

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TEST METHOD NO. 18

WHEEL TORQUE TRANSFER

1. TEST COURSE

An area of asphalt or concrete pavement large enough to accommodate the snowplow, and including a deadman or other type of anchor capable of resisting the maximum drawbar pull of the carrier.

2. TEST APPARATUS

(a) Tire pressure gage.
(b) Electronic load cell with amplifier and recorder, or equivalent means for measuring drawbar pull.
(c) Stop watch.
(d) Lubricant (soap, oil, grease) to eliminate traction on selected wheels.
(e) Cable.

3. TEST PROCEDURES

(a) Check and adjust pressure in all tires to recommended psi.
(b) Connect load cell, cable, anchor point and carrier pintle.
(c) Conduct the test as follows:

(1) Phase 1. The carrier shall be positioned on a smooth, asphalt or concrete pavement and connected, by the rear pintle, through a drawbar dynamometer to an immovable object to prevent forward motion. The drawbar pull with the carrier engine at wide-open-throttle shall be recorded for each forward gear ratio.
(2) Phase 2. Effective traction shall then be denied to three of the carrier wheels by applying a lubricant (soap, grease, or heavy oil) to the pavement under these wheels or to skid plates placed under them. The drawbar pull at wide-open-throttle shall again be measured for each gear ratio. This procedure shall be repeated to measure the drawbar pull of each one of the four wheels.
(3) Phase 3. During each phase of the driveline tests above, the maximum dynamometer reading shall be recorded. The wide-open-throttle application of power in each gear shall be not less than 20 seconds in phase 1 and 10 seconds in phase 2.

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TEST METHOD NO. 19

FUNCTIONAL

1. TEST EQUIPMENT

(a) Complete snowplow serviced and ready for operation.
(b) Electronic load cell with amplifier and recorder (alternate approved equipment may be used).
(c) Towing vehicle.
(d) Spring scale.

2. TEST COURSE

(a) Level road or track with improved or paved surface.

3. TEST PROCEDURE

Start the carrier and plow engines and perform the following operations:

(a) Raise and start plow.
(b) Lower and stop plow.
(c) Operate the discharge chute through its entire range in each direction.
(d) With plow in travel position, drive the snowplow for not less than 2 miles over the road or track, shifting progressively from reverse to low to highest range and from highest to low to reverse for not less than four cycles. During this driving operation bring the snowplow to a full stop, from the maximum speed in high gear, with the service brakes not less than three times.
(e) With plow in travel position, operate the snowplow to make a turn to the right of not less than 270 degrees while traveling forward in low gear.
(f) Repeat (e) above except the 270 degree turn shall be made to the left.
(g) Repeat (e), and (f) above, except reverse gear shall be used.
(h) Stop the snowplow on a level concrete surface with plow off the surface. Apply parking brake and place transmission in neutral. Measure the force required to set the brake. Exert a force equal

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to 50 percent of snowplow weight at the front of the snowplow in a direction parallel to the longitudinal axis of the snowplow and away from the snowplow. Repeat the test with the force being exerted at the rear of the snowplow.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-S-52699B(ME) Snow-Removal Unit, Rotary; Truck-Mounted, Diesel-Engine-Driven, 1500 Tons-Per-Hour Capacity

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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